The Team

BUILDING CLEANCORE

> *"Building waste systems that listen, learn, and respond — for cities that can't wait."*

— MADE BY

AN **Ayesha Noman**
FINAL YEAR PROJECT

AN **Ayesha Nadeem**
FINAL YEAR PROJECT

— SUPERVISED BY

MR **Ms Maha Rasheed**
PROJECT SUPERVISOR

STAY CONNECTED

See It Live. Reach the Team.

DEMO VIDEO

Watch CleanCore in Action

YouTube

CONNECT

Ayesha Noman

LinkedIn

CONNECT

Ayesha Nadeem

LinkedIn

Scan with your phone camera

COMSATS University Islamabad
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CleanCore

Modernizing Smart City Waste Architecture

An AI-driven platform that replaces fixed-schedule waste collection with real-time bin intelligence and optimized routing.

- AI Vision
- Smart Routing
- Live Sync
- Worker Dispatch

A FYP SHOWCASE · COMSATS

INDUSTRY PARTNER
SmartEnds Pvt. Ltd.

01 · THE PROBLEM

Cities Collect on a Hunch.

Most urban waste systems still run on **fixed weekly schedules**. Trucks roll out whether bins are empty or overflowing — wasting fuel on half-full stops and missing the ones that needed pickup yesterday.

The result: streets piled with garbage, citizens frustrated, and municipalities burning kilometers and CO_2 on routes that no longer reflect reality.

~40%
of pickups happen when bins are still half-empty

- **Bins overflow before pickup**
  Health hazards, complaints, fines.
- **Wasted kilometers**
  Half-empty bins still get visited.
- **No real-time visibility**
  Admins and drivers operate blind.
- **Manual dispatching**
  Critical bins fall through the cracks.

> *A modern city cannot run on yesterday's schedule. CleanCore listens to bins in real time and dispatches accordingly.*
>
> — DESIGN PRINCIPLE

THE SAME ROUTE HITS ALL THREE

02 · OUR SOLUTION

Replace the Schedule with Intelligence.

CleanCore turns a static collection routine into a live, AI-driven operation. Cameras read fill levels, the system reroutes drivers in real time, and every kilometer is justified.

01
AI Bin Vision
YOLOv8 reads fill level & waste type from images.

02
Smart Routing
Stops sequenced for shortest drive — less fuel, less CO_2.

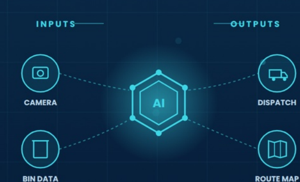
03
Live Bin Sync
Admin & drivers see the same status — no refresh.

04
Worker Dispatch
Critical bins land on the right driver instantly.

WHY IT MATTERS

No new hardware retrofit — works with existing cameras. Lower fuel costs, lower emissions, faster response to overflow incidents.

↓ 35% LESS FUEL	↓ 40% FEWER KMS
100% LIVE SYNC	0 HARDWARE RETROFIT



AI CORE ORCHESTRATES THE LOOP

03 · SYSTEM FLOW

How a Bin Becomes a Route.

1. **IoT Camera** — Bin-mounted camera captures images at intervals. (01)
2. **YOLOv8 Model** — Detects fill level and waste category from frames. (02)
3. **Firebase** — Real-time database syncs every bin instantly. (03)
4. **App & Web** — Worker & admin interfaces — live, no polling. (04)
5. **OSRM Routing** — Sequences stops for the shortest collection drive. (05)

TECH STACK

YOLOv8 · Firebase · Flutter · React · OSRM · Python · Node.js

END-TO-END

From a single bin photo to a re-sequenced driver route in **under 2 seconds** — every step on infrastructure that's already familiar to engineering teams.

LIVE ROUTE ACROSS THE CITY